

Rahul Bhushan Vemula

AI Content Engineer | Java Full Stack Developer | AI & Data Science Graduate

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SUMMARY

AI & Data Science graduate and AI Content Engineer at StudyEasy.org, building production-grade full-stack applications and integrating applied machine learning into real-world content pipelines. Comfortable across the stack — from Spring Boot backends to React/Angular frontends to model evaluation and deployment. Open to roles in full-stack development, applied AI, and machine learning engineering.

EXPERIENCE

AI Content Engineer — StudyEasy.org

May 2026 – Present

Remote

- **Building AI content pipelines** for DubNext.com, StudyEasy's AI-powered learning management system — chaining LLMs, voice, video, and translation tools into automated course-generation workflows.
- Research and **evaluate AI tools** (language, voice, video, translation models) against a quality bar before integrating them into production teaching workflows.
- Quality-check generated output and contribute to backend and frontend development across the platform where needed.

AI & Machine Learning Intern — EduSkills × Google for Developers

Oct 2025 – Dec 2025

Virtual — Google for Developers India Edu Program (10 weeks)

- Applied core AI/ML concepts: supervised and unsupervised learning, model training, and evaluation on real-world datasets.
- Worked through the end-to-end ML workflow — data preprocessing, feature engineering, model selection, and performance analysis.

Java Full Stack Development Intern — IIDT Blackbucks

May 2025 – Jul 2025

Virtual

- Built user-centric features end-to-end — login/registration and secure CRUD operations — using Java, Spring Boot, HTML, CSS, JavaScript, and React.
- Contributed across the frontend and backend boundary on real application flows.

Machine Learning Intern — NIELIT

Jun 2023 – Aug 2023

Virtual

- Worked through the full ML lifecycle — data collection, feature engineering, training, and deployment — using Python, Pandas, and Matplotlib.

PROJECTS

Anomaly-Based Video Forgery Detection (Deepfake Detection)

2025 – 2026

Python, TensorFlow, OpenCV, Flask — Final-Year Thesis Project

- Built a deepfake detection system using a Convolutional Autoencoder with LSTM, trained on real videos only — framing detection as anomaly detection so it generalises to unseen manipulation methods.
- Captured both spatial and temporal forgery cues; deployed as a Flask web app with secure login and prediction history.

Online Banking System

Jun 2024

Java, Spring Boot, Spring Security, JWT, React.js, MySQL

- Designed and built a secure banking application end-to-end — account creation, fund transfers, deposits, withdrawals, and transaction history.
- Implemented JWT-based role authentication for separate customer and admin dashboards; persisted accounts, transactions, and audit logs in MySQL.

Hospital Management Platform

Jun 2025

Java, Spring Boot, Angular, Spring Security, MySQL

- Built a platform streamlining patient, doctor, and admin workflows — registration, appointment scheduling, and electronic health records.
- Enforced role-based access via Spring Security across all roles, fronted by an Angular UI.

MediTrack — SMS Notification System

Mar 2025

Java, Spring Boot, Twilio API, REST APIs

- Developed a Spring Boot backend delivering automated SMS alerts via the Twilio API, with a verification flow ensuring valid, confirmed recipients.
- Designed for healthcare use cases — appointment reminders, medicine alerts, and emergency notifications.

Facial Recognition Attendance System

Dec 2025

Python, OpenCV, LBPH, MySQL, MongoDB

- Built a real-time face-recognition system that automates attendance marking using the LBPH algorithm on OpenCV.
- Used hybrid storage — MySQL for attendance records, MongoDB for user profiles and media.

EDUCATION

SRKR Engineering College, Bhimavaram

Nov 2022 – Apr 2026

B.E. in Artificial Intelligence & Data Science — CGPA: 8.36 / 10

Relevant Coursework: AI & ML, Data Science, Object-Oriented Programming, Data Structures & Algorithms

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, R, HTML/CSS

Frameworks & Backend: Spring Boot, Spring Security, Hibernate/JPA, REST APIs, React.js, Angular, Bootstrap

Data & ML: TensorFlow/Keras, scikit-learn, Pandas, NumPy, Matplotlib/Seaborn, OpenCV, Flask

Databases: MySQL, PostgreSQL, MongoDB, SQLite

Tools & Cloud: Git/GitHub, Docker, Kubernetes, AWS, Postman, Maven, IntelliJ, VS Code

CERTIFICATIONS

- Full Stack Java Developer — Java, JSP, Spring Boot, JavaScript, React (Udemy)
- Artificial Intelligence & Machine Learning (Udemy)
- Programming with R (NPTEL)